EA : Statistical Learning Theory

Analysis of a research paper (English version): Weighted Nadaraya-Watson Regression Estimation, Zongwu Cai

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November 21, 2022

Paper Framework

- We consider a dataset of size n: (X_i, Y_i) i.i.d.
- And we try to estimate with $\hat{m}(x)$ the "true regression function" $m(x) = \mathbb{E}[Y_i | X_i = x]$
- In the lectures, we used Stone theorem [Stone 1997, Consistent Nonparametric Regression] with sufficient conditions to get:

$$\mathbb{E}\left[(r_n(X) - r(X))^2\right] \longrightarrow 0$$

- Here, we chose a point x in the interior of our interval : we study $\hat{m}(x) - m(x)$ and the behaviour of $\hat{m}(ch_n)$ with $h_n \longrightarrow 0$ et $c \in]0, 1[$ (boundary behaviour if the marginal density of X is $[0, 1]$).

Paper framework

- Non parametric kernel estimator, special case of local average estimator
- Kernal : positive real function K, absolutely integrable. Often normalized, continuous (probability distribution density) and symetrical ((cf. theorem 1 hypothesis).

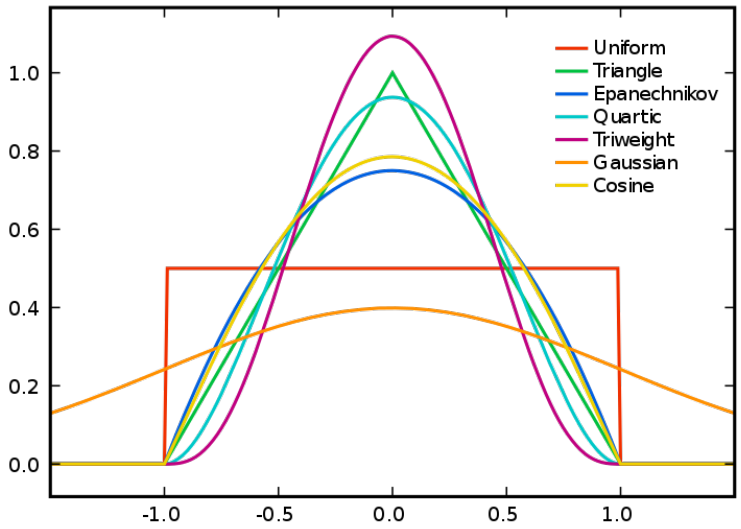


Figure: Kernel examples

Kernel and bandwidth parameter

- General kernel estimator formula (with weights)

$$\hat{m}(x) = \sum_{i=1}^{n} c_{n,i}(x, \mathcal{D}) K_{h,n}(x - X_i) Y_i$$

with $K_{h,n}$ kernels, h a bandwidth parameter and $c_{n,i}$ weights.

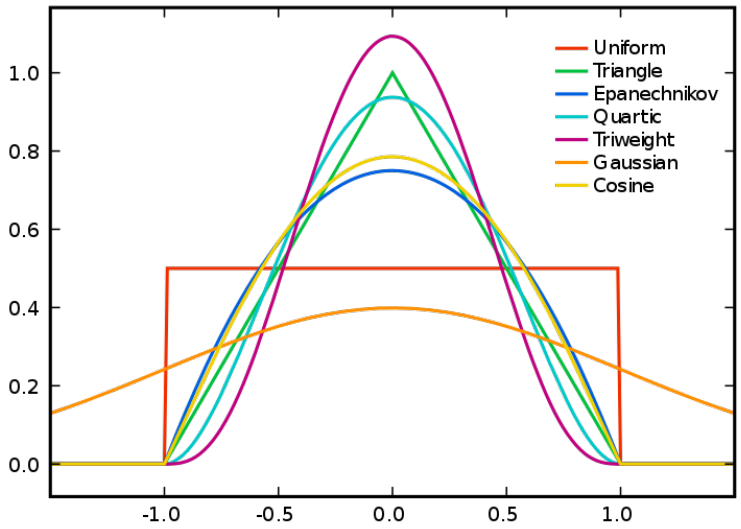


Figure: Kernel examples

Some well-known estimators

Suppose $\mathbb{E}[|Y_i|] < \infty$ and let $K_h(x) = \frac{1}{h}K(\frac{x}{h})$.

- Nadaraya Watson estimator:

$$\hat{m}_{NW}(x) = \frac{\sum_{i=1}^{n} K_h(x - X_i)Y_i}{\sum_{i=1}^{n} K_h(x - X_i)}$$

- Local Linear estimator:

$$\hat{m}_{LL}(x) = \sum_{i=1}^{n} w_i(x)Y_i \quad \text{où} \quad w_i(x) = \frac{K_h(x - X_i)(S_{n,2} - (X_i - x)S_{n,1})}{S_{n,0}S_{n,2} - S_{n,1}^2}$$

avec

$$S_{n,j} = \sum_{i=1}^{n} K_h(x - X_i)(X_i - x)^j$$

(Obtained by solving a locally weighted linear regression)

Goal of paper

- Weighted Nadaraya-Watson estimator (WNW)

$$\hat{m}_{WNW}(x) = \frac{\sum_{i=1}^{n} w_i(x) K_h(x - X_i) Y_i}{\sum_{i=1}^{n} w_i(x) K_h(x - X_i)}$$

- with w_i weights, function of data $X_1, \ldots, X_n$, positive and so that$\sum_{i=1}^{n} w_i(x) = 1$, and

$$\sum_{i=1}^{n} (X_i - x) w_i(x) K_h(x - X_i) = 0 \tag{1}$$

- The formula for the w_i, obtained by an optimisation methods :

$$w_i(x) = n^{-1}\{1 + \lambda(X_i - x) K_h(x - X_i)\}^{-1}$$

- with λ (a function of lagrange multipliers from the optimisation problem), chosen to minimize

$$L_n(\lambda) = \frac{1}{nh} \sum_{i=1}^{n} \log\left(1 + \lambda(X_i - x) K_h(x - X_i)\right)$$

First result

- Studying bias, variance (and MSE) of the WNW estimator at a point x.

Theorem 1

Theorem 1 Suppose that conditions B1-B4 (next slide) are met then, when $n \to \infty$:

- *First :*

$$\hat{m}(x) - m(x) = \frac{1}{2} h^2 \mu_2 m''(x) + o_p(h^2) + \mathbf{O}_p(((nh)^{-\frac{1}{2}}))$$

- *Then :*

$$\sqrt{nh}\left[\hat{m}(x) - m(x) - \frac{1}{2} h^2 \mu_2 m''(x) + o_p(h^2)\right] \xrightarrow{\mathcal{L}} \mathcal{N}(0, \nu_0 \sigma^2(x)/g(x))$$

Conditions (simplified given the stronger independent setting we use for the presentation)

Let $g(.)$ be the marginal density of X_i. and let $\mu_j = \int u^j K(u)du$, $\nu_j = \int u^j K^2(u)du$, and $\sigma^2(x) = Var(Y_i|X_i = x)$.

- B1 :
  $\forall x$, on a $g(x) > 0$, $g(.)$ and $\sigma^2(.)$ are continuous at x,with $m(.)$twice continuously differentiable in a neighborhood of x.
- B2 :
  K(.) is a bouned, symetric density with compact support included in $[-1, 1]$.
- B3 :
  $g_{Y|X} \leqslant M_1 < \infty$ where $g_{Y|X}$ is the conditional density of Y given X.
- B4 : $\mathbb{E}(|Y_i|^\delta|X_i = u) \leqslant M_3 < \infty$ in a neighborhood of x,for some $\delta > 2$

Idea of proof for theorem 1

How to begin?

First compute $\hat{m}(x) - m(x)$ for some x then 3 terms appear (out of nowhere I admit).

$$\begin{aligned}\hat{m}(x) - m(x) &= \frac{\sum_{i=1}^{n}[Y_i - m(x)]w_i(x)K_h(x - X_i)}{\sum_{i=1}^{n} w_i(x)K_h(x - X_i)} \\ &= \{(nh)^{-\frac{1}{2}}J_1 + J_2\}J_3{}^{-1}\{1 + o_p(1)\}\end{aligned}$$

For all of these 3 terms (defined later) we will compute their expected value, then use a version of the Law of Large Numbers and finally some dominated convergence with a healthy dose of central limit theorem!

Elements of proof for theorem 1

Let $\epsilon_i = Y_i - m(X_i)$, and

$$b_i(x) = \left[1 - \frac{h\mu_2 g'(x)}{2\nu_2 g(x)}(X_i - x)K_h(x - X_i)\right]^{-1}$$

Let $\xi_i = \sqrt{h} b_i(x)\epsilon_i K_h(x - X_i)$.

Lemma 1

If conditions B1 to B4 are verified then :

$$Var(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\xi_i) \longrightarrow \nu_0\sigma^2(x)g(x)$$

Proof of lemma 1

Firstly, $\mathbb{E}[\xi_i] = \mathbb{E}[\mathbb{E}[\xi_i|X_i]] = \mathbb{E}[\sqrt{h}b_i(x)K_h(x - X_i)\mathbb{E}[\epsilon_i|X_i]]$

and $\mathbb{E}[Y_i|X_i] = m(X_i)$ so $\mathbb{E}[\epsilon_i|X_i] = 0$ and $\mathbb{E}[\xi_i] = 0$

Because (X_i) are independents, (ξ_i) are too and we can write
$\mathrm{Var}(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\xi_i) = \mathrm{Var}(\xi_i) = \mathbb{E}[{\xi_i}^2] = h\mathbb{E}[{b_i}^2(x){\xi_i}^2{K_h}^2(x - X_i)]$

An explicit calculation of the expected value and use of the dominated convergence theorem finishes the proof.

Lemma 2

As a reminder, λ was chosen (slide number 6) so that to minimize $L_n(\lambda) = \frac{1}{nh}\sum_{i=1}^{n}\log\{1+\lambda(X_i - x)K_h(x - X_i))\}$

Lemma 2

If conditions B1-B4 are verified then :

$$\lambda = -\frac{h\mu_2 g'(x)}{2\nu_2 g(x)}\{1 + o_p(1)\}$$

And furthermore, $w_i(x) = n^{-1}b_i(x)\{1 + o_p(1)\}$

Proof of Lemma 2

$\forall j \geq 1$, let $A_j = \frac{1}{n}\sum_{i=1}^{n}(X_i - x)^j K_h{}^j(x - X_i)$

Using the central limit theorem and dominated convergence theorem, $A_1 = -\frac{1}{2}\mu_2 h^2 g'(x) + o_p(h^2)$, $A_2 = h\nu_2 g(x) + o_p(h^2)$, and $A_3 = \mathbf{O}_p(h^2)$ (We humbly suggest there might be a typo in the paper and no $1/2$ factor for A_1 but this does not affect fundamentaly the final consistency result

Chen and Hall (1993) gives $|\lambda| = \mathbf{O}_p(h)$. Then, by using $L_n'(\lambda) = 0$ we have :

$0 = L_n'(\lambda) = A_1 - \lambda A_2 + \lambda^2 A_3 + \mathbf{O}_p(\lambda^3)$, then :

$\lambda = \frac{A_1}{A_2} + \lambda^2 \frac{A_3}{A_2} + \mathbf{O}_p(\lambda^3) = -\frac{h\mu_2 g'(x)}{2\nu_2 g(x)}\{1 + o_p(1)\}$
From what we can write
$w_i(x) = \frac{1}{n}\left[1 - (\frac{h\mu_2 g'(x)}{2\nu_2 g(x)}\{1 + o_p(1)\})(X_i - x)K_h(x - X_i)\right]^{-1}$
and $w_i(x) = n^{-1} b_i(x)\{1 + o_p(1)\}$

Final proof elements for Theorem 1

Let $J_2 = \frac{1}{n}\sum_{i=1}^{n}[Y_i - m(x)]b_i(x)K_h(x - X_i)$
and $J_3 = \sum_{i=1}^{n} b_i(x)K_h(x - X_i)$

$$\hat{m}(x) - m(x) = \frac{\sum_{i=1}^{n}[Y_i - m(x)]w_i(x)K_h(x - X_i)}{\sum_{i=1}^{n} w_i(x)K_h(x - X_i)}$$
$$= \{(nh)^{-\frac{1}{2}}J_1 + J_2\}J_3{}^{-1}\{1 + o_p(1)\}$$

ANd (using Taylor and B1) :

$\mathrm{J}_2 = \frac{1}{2n}\sum_{i=1}^{n} m''(x)(X_i - x)^2 b_i(x)K_h(x - X_i) + o_p(h^2)$
$= \frac{h^2}{2}\mu_2 m''(x)g(x) + o_p(h^2)$

and finally,

$$J_3 = g(x) + o_p(1)$$

Final proof elements for Theorem 1

Therefore,
$\sqrt{nh}\left[\hat{m}(x) - m(x) - \frac{1}{2}h^2\mu_2 m''(x) + o_p(h^2)\right] = g^{-1}(x)J_1 + o_p(1)$

The central limit theorem allows us to conclude using lemma 1
$Var(J_1) \longrightarrow \nu_0\sigma^2(x)g(x)$:

$$\sqrt{nh}\left[\hat{m}(x) - m(x) - \frac{1}{2}h^2\mu_2 m''(x) + o_p(h^2)\right] \xrightarrow{\mathcal{L}} \mathcal{N}(0, \nu_0\sigma^2(x)/g(x))$$

Consequence of theorem 1

- Significant amelioration on Nadaray-Watson !

	Bias	Variance	Asymptotic minimax efficiency
LL	$\frac{1}{2}m''(x)\mu_2 h^2$	$\frac{\sigma^2(x)}{g(x)nh}\nu_0$	100
NW	$\frac{1}{2}[m''(x) + \frac{2m'(x)g'(x)}{g(x)}]h^2\mu_2$	$\frac{\sigma^2(x)}{g(x)nh}\nu_0$	0
WNW	$\frac{1}{2}m''(x)\mu_2 h^2$	$\frac{\sigma^2(x)}{g(x)nh}\nu_0$	nearly 100

with $\mu_2 = \int u^2 K(u)du$, $\nu_0 = \int K^2(u)du$ et g the marginal density X_i.

Convergence speed

- All the estimators have an MSE of this form: $Ah^4 + \frac{B}{nh}$.
- There exists for all n and at each point x, an optimal h_n
- We have

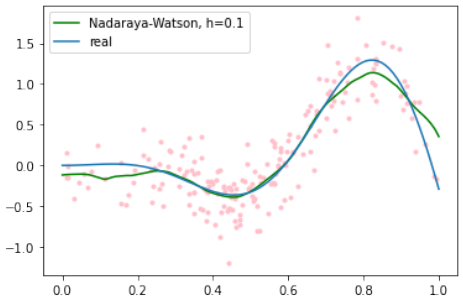
$$h_{\text{opt}} = \left(\frac{\nu_0 \sigma^2(x)}{\mu_2^2 g(x) \left\{ m''(x) \right\}^2} \right)^{1/5} n^{-1/5}$$

  and

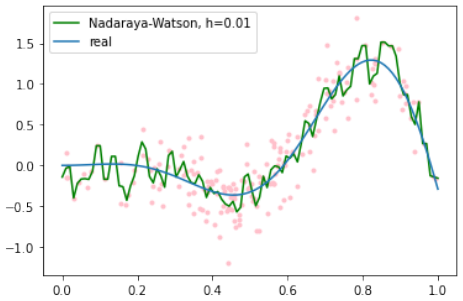
$$\text{MSE}_{opt} = \frac{5}{4} \left(\frac{\sqrt{\mu_2 \mid m''(x) \mid} \nu_0 \sigma^2(x)}{ng(x)} \right)^{4/5}$$

  for both WNW and LL estimator
- With $_{opt}$ et h_{opt}, we can compute the asymptotic minimax efficiency.

Choice of K and h for K_h (see attached notebook)



(a)



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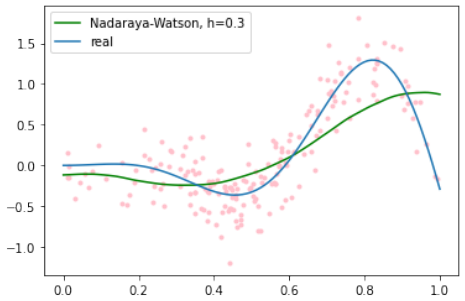
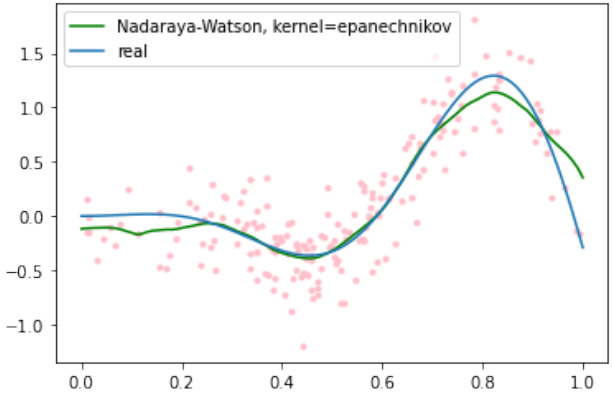


(c)

Figure: $h = 0.1$, $h = 0.01$, $h = 0.3$

Choice for h is a classical bias-variance trade-off with an emphasis on the role of convexity in high bias (low density g is a classical cause of variance)

Choice of K and h for K_h



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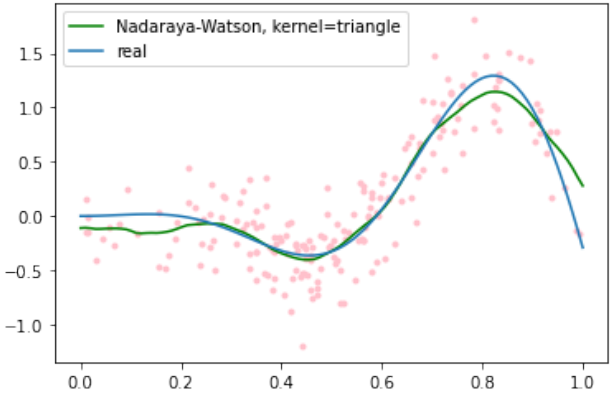

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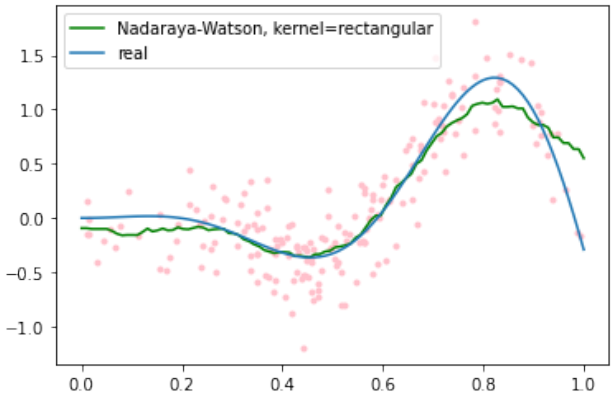

(c)

Figure: Epanechnikov, triangular or rectangular kernel

Remark on consitency of a NW-like estimator

With theorem 1, it is clear that the two following hypothesis are needed to ensure consistency overall of a NW-like estimator:

- $h \longrightarrow 0$
- $nh_n \longrightarrow +\infty$

But these two conditions alone are not sufficient to ensure overall consistency. To apply [Stone 1997], we have to add uniform continuity of g. Alternatively, we could make hypothesis on two functions: m'' and σ^2.

Second result

- Hypothesis : K has compact support $[-1, 1]$ and (additionaly in comparison to Theorem 1) g also has compact support $[0, 1]$.
- Study the boundary behaviour at 0^+ . Let c be a constants speed parameter with $0 < c < 1$

Theorem 2

Suppose that B1 to B4 are met. then, for $n \to \infty$:

$$\sqrt{nh}\left[\hat{m}(ch) - m(ch) - B_c(h) + o_p(h^2)\right] \xrightarrow{\mathcal{L}} \mathcal{N}(0, {\sigma_c}^2)$$

with $B_c(h)$ and σ_c^2 that give for $c > 1/2$ a similar MSE for NWN and LL and a better behaviour for LL when $c < 1/2$.
Note that even if bias can be smaller at the boundary than in the interior for some kernel, it is possible to show that MSE will always be worse suggesting higher variance at the boundary.

Conclusion

- Similar performance of the WNW estimator with a lower computational complexity and faster runtime(see the implementation).
- Used in economy and finance. Problem: data is not iid, usually a stationnary markov chain. Need to use α-mixing for the proofs with additional hypothesis.
- A few limitation...Kernel with compact support so no gaussian kernel and $m \in \mathcal{C}^2$.

Annex

Minimax efficiency

- Linear minimax risk is defined by

$$R(n, \mathcal{C}_2) = \inf_{\hat{S} linear} \sup_{m \in \mathcal{C}_2} E\left[(\hat{S} - m(x_0))^2 | \mathcal{F}_n^X\right]$$

(Linear does not mean linear regressor but linear in (Y_i)

- Minimax Afficiency of a local average estimator $\hat{m}_L$ is then defined by

$$\left(\frac{R(n, \mathcal{C}_2)}{\sup_{m \in \mathcal{C}_2} E[(\hat{m}_L(x_0) - m(x_0))^2 | \mathcal{F}_n^X]}\right)^{\frac{5}{4}}$$